

PROBLEM EXISTING

- **Micro-radian aim** : A laser link is hair-thin. Both terminals must stay pointed at each other within **micro-radians** while vehicles, drones and masts move and shake. (Kaymak et al., IEEE Comm. Surveys & Tutorials, 2018)
- **Wrong-target lock** : Stars, sun glints, lamps and reflections look like the beacon. A **brightness-only tracker locks onto the wrong light** and the link never closes.
- **Delay & dropouts** : Camera + command **latency (≈ 40 ms)** and blink / scintillation gaps make simple trackers lag and drop the beam, forcing a slow manual re-alignment.

PROPOSED SOLUTION

We propose ZeroDrift, a software coarse-alignment system that detects the partner's 4 Hz beacon in a camera image, proves it is the right light, and steers the terminal to keep it centred: solving PS26169 without special hardware.

- **Physics-true virtual camera** : real orbits (SGP4), turbulence, vibration, decoys and sensor noise, so every algorithm is **tested before any hardware exists**.
- **Detect** : top-hat + matched filter + CFAR + sub-pixel centroid finds a **faint beacon in a cluttered sky**.
- **Identify** : 4 Hz blink test + neural verifier give each light an identity: **0 / 64 wrong-target locks**.
- **Track & coast** : IMM Kalman filter holds the target through dark half-cycles: **98.3 % lock held** (featured demo run).
- **Control** : Smith predictor cancels the command delay: **median error 198 μ rad**, lock in **1.27 s** (median, 64 runs).

UVP (UNIQUE VALUE PROPOSITION)**Identity, not brightness**

Locks only on the beacon's blink signature; lamps, stars and wrong-rate strobes are vetoed.

Delay-cancelled closed loop

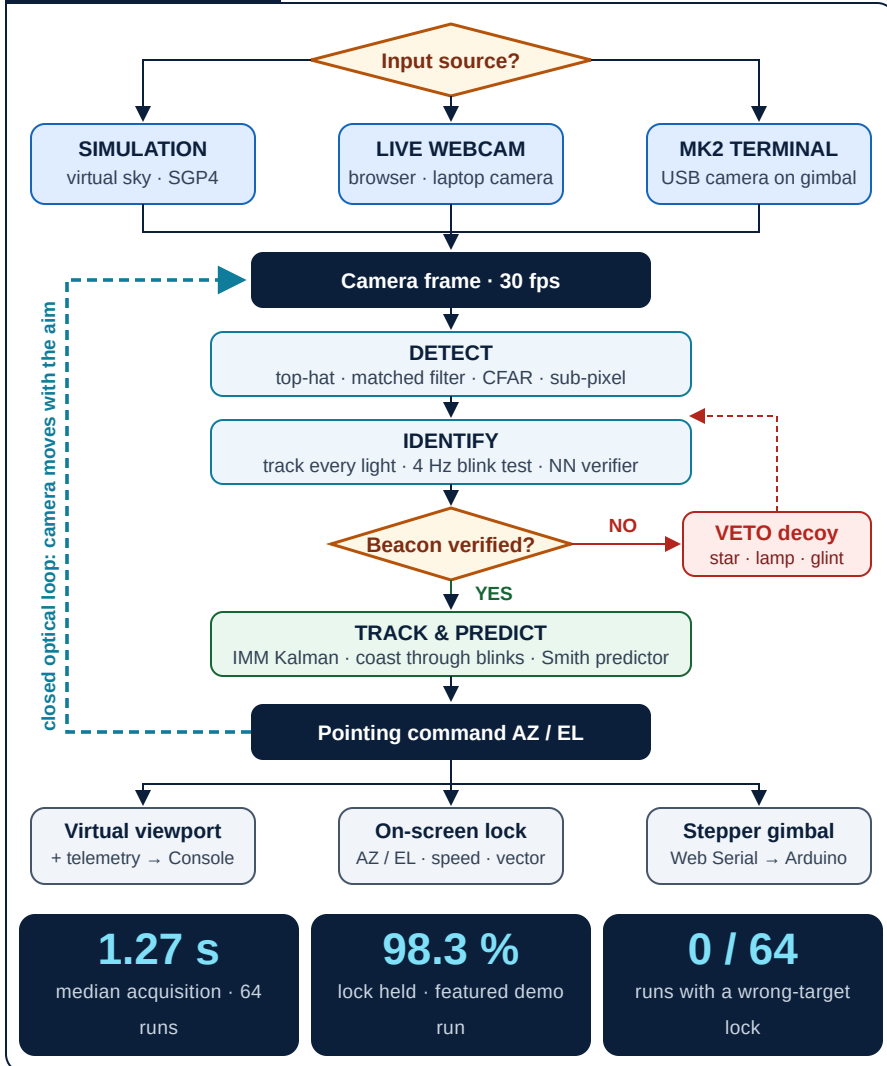
The camera rides the gimbal, so error is measured on the image itself; the Smith predictor removes lag.

One engine, three modes

Same pipeline runs in simulation, on a laptop webcam, and on our MK2 stepper terminal.

Measured, not claimed

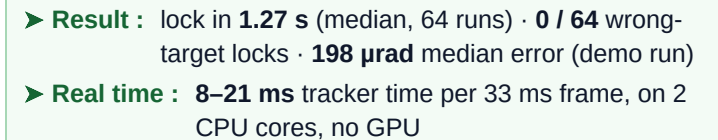
Open source, 250 passing automated tests; every number on these slides regenerates from logged runs.

ARCHITECTURE

SOFTWARE FLOWCHART · IMPLEMENTED IN src/fsoc_pat (pipeline · detection · tracking · control)



PROOF > PROMISE · WORKING PROTOTYPE



METHODOLOGY

1 Model the physics → **2** build each stage with a unit test → **3** 64-run Monte-Carlo campaign → **4** live webcam demo → **5** MK1 / MK2 hardware in the loop.